177915: metasandstone, Randalls Mine

Location and sampling

WIDGIEMOOLTHA (SH 51-14), MOUNT BELCHES (3335)
MGA Zone 51, 425113E 6564884N

Sampled on 3 May 2004

The sample was taken from a 1 m-diameter block on the mullock heap at the abandoned Randalls gold mine.

Tectonic unit/relations

The unit sampled is a dark grey, medium-grained and metamorphosed volcaniclastic sandstone of the Mount Belches Formation, on the northern margin of the Randall Dome (Painter and Groenewald, 2001), within the Bulong Domain of the Kurnalpi Terrane, Eastern Goldfields Superterrane, Yilgarn Craton (Cassidy et al., 2006). At this locality, the Mount Belches Formation contains a bedding-parallel foliation that predates the development of upright, northwest-trending macroscopic folds attributed to regional D_2. The rocks are also affected by post-D_2 contact metamorphism, as a result of the emplacement of granitic rocks into the Randall Dome (Painter and Groenewald, 2001). It is possible that the Kiaki Monzogranite (GSWA 177914; Wingate and Bodorkos, 2007) is representative of these late-stage granitic rocks.

Petrographic description

The visually estimated primary mineralogy of this sample includes 20% sand-sized quartz grains, 15% sand-sized plagioclase grains, 10–15% fine-grained biotite, 55% fine-grained quartzofeldspathic matrix, and accessory chlorite and carbonate minerals. Abundant single-crystal quartz and plagioclase grains are up to 1 mm long. Fine biotite is disseminated as patches, short parallel lamellae, and as isolated flakes up to 0.3 mm long. Chlorite grains up to 0.5 mm in size are sparsely disseminated and are not obviously related to biotite. Rare carbonate minerals, apatite, opaque oxide grains, and leucoxene are also disseminated.

The larger quartz and feldspar grains are rounded to subrounded, and define a weak layering. They are interpreted to be of relict detrital origin, with the layering being parallel to bedding. A weak foliation is evident in the recrystallized matrix, although individual biotite and chlorite crystals are unoriented. The texture and mineralogy is consistent with static recrystallization during contact metamorphism, probably under lower-greenschist facies conditions. Minor clay clouding in some of the feldspars probably reflects incipient weathering.

Zircon morphology

The zircons isolated from this sample are subhedral to euhedral, and range from clear and colourless to mid-brown. They are up to 300 µm long, with aspect ratios up to 5:1. Concentric growth zoning is ubiquitous, and a few zircons show disrupted internal structures, consistent with alteration. Cathodoluminescence images of representative zircons are shown in Figure 1.

Analytical details

This sample was analysed on 26–27 February 2005, using SHRIMP-B. Fourteen analyses of the CZ3 standard indicated an external spot-to-spot (reproducibility) uncertainty of 1.68% (1σ), and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.49% (1σ). Common-Pb corrections were applied to all analyses using contemporaneous common-Pb isotopic compositions determined by the method of Stacey and Kramers (1975).

Results

Sixty-seven analyses were obtained from 66 zircons, with one grain (9) analysed twice. Results are listed in Table 1, and shown in a concordia diagram (Fig. 2) and a probability density diagram (Fig. 3).

Interpretation

Most analyses are concordant to slightly discordant (Fig. 2), although three analyses are characterized by moderate to strong discordance (>10%). The dates obtained from these three analyses (Group D; Table 1) are imprecise or unreliable, and are not considered geologically significant. The remaining 64 analyses can be divided into three groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios and their uranium contents.

Group 1 comprises six analyses of six zircons (Table 1) with moderate uranium contents (103–449 ppm), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2660 ± 14 Ma (MSWD = 3.02).

Group 2 comprises 49 analyses of 49 zircons (Table 1) with widely varying uranium contents (16–797 ppm), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 3460–2671 Ma.

Group 3 comprises nine analyses of eight zircons (Table 1) with very high uranium contents (880–8416 ppm), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2207–1745 Ma.

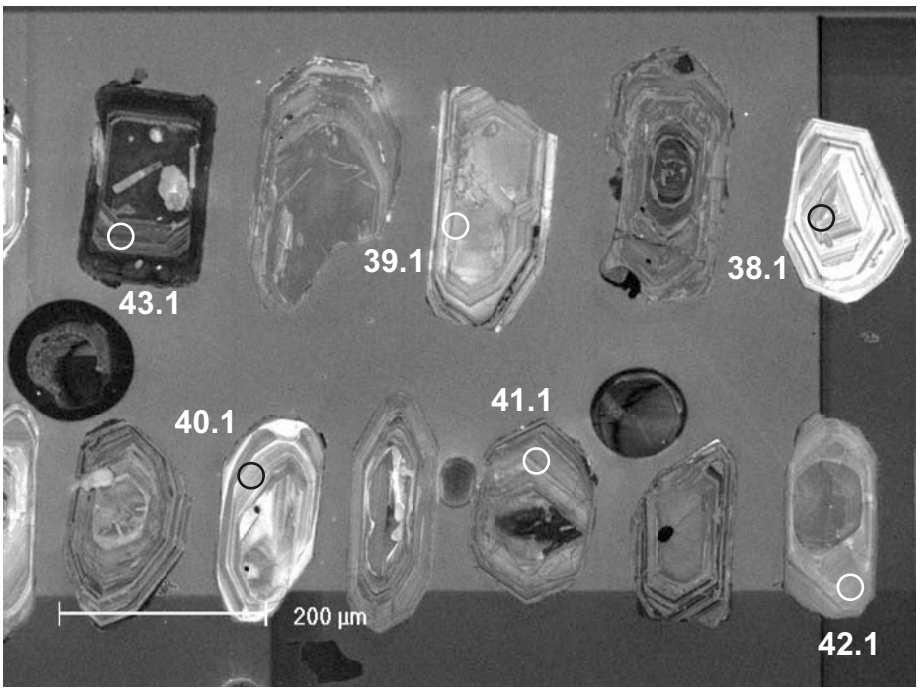

Figure 1. Cathodoluminescence image of representative zircons from sample 177915: metasandstone, Randalls Mine. Numbered circles indicate approximate positions of analysis sites

If all of the analyses in Group 1 are of unmodified detrital zircons, then the date of 2641 ± 17 Ma (1σ) for analysis 66.1 represents a possible maximum age for deposition of the sandstone protolith of this metasedimentary rock. However, a more robust estimate of the maximum depositional age is 2660 ± 14 Ma, based on the weighted mean $^{207}Pb^{*}/^{206}Pb^{*}$ date of the six zircons in Group 1.

Considered together with the six analyses in Group 1, the 49 analyses in Group 2 define a major age component at c. 2676 Ma, and several minor components spanning the range 3460–2780 Ma (Fig. 3). These are interpreted as the ages of zircon-bearing rocks in the detrital source region(s) of the sandstone protolith.

The nine analyses in Group 3 indicate a wide range of younger $^{207}Pb^{*}/^{206}Pb^{*}$ dates, which display a well-defined negative correlation with uranium contents over the range 880–8416 ppm. These results are interpreted to reflect one or more Paleoproterozoic episodes of radiogenic-Pb loss from these sites, possibly facilitated by radiation damage to the zircon lattices as a consequence of their elevated uranium contents, and the dates are not considered to be geologically significant.

References

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Recommended reference for this publication

WINGATE, M. T. D., and BODORKOS, S., 2007, 177915: metasandstone, Randalls Mine; Geochronology dataset 664, *in* Compilation of geochronology data: Western Australia Geological Survey.

Data obtained: 27 February 2005
Data released: 31 May 2007

Table 1. Ion microprobe analytical results for zircons from sample 177915: metasandstone, Randalls Mine

Grp no.	Spot no.	Grain .spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ±1σ	$^{207}Pb/^{206}Pb$ ±1σ	$^{238}U/^{206}Pb^*$ ±1σ	$^{207}Pb^*/^{206}Pb^*$ ±1σ	$^{238}U/^{206}Pb^*$ date (Ma) ±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ	Disc (%)
1	66	66.1	103	62	0.63	0.025	2.002 ± 0.040	0.17897 ± 0.00175	2.003 ± 0.040	0.17875 ± 0.00180	2611 ± 43	2641 ± 17	1.2
1	64	64.1	416	184	0.46	0.074	2.050 ± 0.036	0.17963 ± 0.00074	2.052 ± 0.036	0.17897 ± 0.00078	2559 ± 37	2643 ± 7	3.2
1	18	18.1	318	120	0.39	0.095	1.987 ± 0.035	0.18016 ± 0.00076	1.989 ± 0.035	0.17931 ± 0.00080	2626 ± 38	2647 ± 7	0.8
1	32	32.1	196	169	0.89	0.013	1.940 ± 0.035	0.18173 ± 0.00095	1.940 ± 0.035	0.18161 ± 0.00095	2679 ± 40	2668 ± 9	–0.4
1	26	26.1	449	355	0.82	0.037	1.966 ± 0.035	0.18206 ± 0.00058	1.967 ± 0.035	0.18174 ± 0.00059	2650 ± 39	2669 ± 5	0.7
1	20	20.1	257	121	0.49	0.026	2.027 ± 0.036	0.18211 ± 0.00079	2.027 ± 0.036	0.18188 ± 0.00083	2585 ± 38	2670 ± 8	3.2
2	36	36.1	622	297	0.49	0.017	2.015 ± 0.035	0.18211 ± 0.00055	2.016 ± 0.035	0.18197 ± 0.00057	2597 ± 37	2671 ± 5	2.8
2	13	13.1	128	56	0.46	–0.055	1.972 ± 0.037	0.18172 ± 0.00113	1.971 ± 0.037	0.18221 ± 0.00116	2646 ± 40	2673 ± 11	1.0
2	19	19.1	132	44	0.35	0.137	2.038 ± 0.041	0.18344 ± 0.00110	2.040 ± 0.041	0.18222 ± 0.00126	2571 ± 43	2673 ± 11	3.8
2	63	63.1	166	98	0.61	–0.088	1.992 ± 0.038	0.18152 ± 0.00112	1.991 ± 0.038	0.18231 ± 0.00124	2624 ± 41	2674 ± 11	1.9
2	46	46.1	310	105	0.35	0.016	1.949 ± 0.035	0.18269 ± 0.00130	1.949 ± 0.035	0.18254 ± 0.00131	2669 ± 39	2676 ± 12	0.3
2	35	35.1	206	85	0.43	–0.016	2.058 ± 0.037	0.18242 ± 0.00093	2.057 ± 0.037	0.18257 ± 0.00096	2554 ± 38	2676 ± 9	4.6
2	62	62.1	124	64	0.54	–0.020	2.024 ± 0.038	0.18262 ± 0.00129	2.024 ± 0.038	0.18279 ± 0.00130	2589 ± 41	2678 ± 12	3.3
2	25	25.1	173	104	0.62	–0.009	2.026 ± 0.037	0.18288 ± 0.00099	2.026 ± 0.037	0.18296 ± 0.00100	2586 ± 39	2680 ± 9	3.5
2	12	12.1	112	70	0.65	–0.114	1.979 ± 0.037	0.18197 ± 0.00122	1.977 ± 0.037	0.18298 ± 0.00129	2639 ± 41	2680 ± 12	1.5
2	11	11.1	54	42	0.80	0.077	2.128 ± 0.046	0.18371 ± 0.00181	2.130 ± 0.046	0.18302 ± 0.00189	2482 ± 44	2680 ± 17	7.4
2	54	54.1	166	124	0.77	0.865	2.052 ± 0.039	0.19083 ± 0.00128	2.070 ± 0.039	0.18311 ± 0.00194	2541 ± 40	2681 ± 18	5.2
2	51	51.1	110	43	0.40	0.016	2.018 ± 0.040	0.18328 ± 0.00147	2.019 ± 0.040	0.18313 ± 0.00152	2594 ± 42	2681 ± 14	3.3
2	31	31.1	139	46	0.34	0.003	1.954 ± 0.036	0.18327 ± 0.00111	1.954 ± 0.036	0.18324 ± 0.00113	2665 ± 40	2682 ± 10	0.7
2	65	65.1	229	107	0.48	–0.038	1.965 ± 0.036	0.18294 ± 0.00105	1.965 ± 0.036	0.18328 ± 0.00109	2652 ± 40	2683 ± 10	1.1
2	38	38.1	102	61	0.62	–0.062	1.971 ± 0.040	0.18273 ± 0.00137	1.969 ± 0.040	0.18328 ± 0.00137	2647 ± 44	2683 ± 12	1.3
2	2	2.1	98	58	0.61	–0.012	2.075 ± 0.040	0.18319 ± 0.00141	2.075 ± 0.040	0.18330 ± 0.00141	2536 ± 41	2683 ± 13	5.5
2	30	30.1	68	45	0.68	–0.066	1.926 ± 0.039	0.18285 ± 0.00163	1.925 ± 0.039	0.18344 ± 0.00167	2697 ± 44	2684 ± 15	–0.5
2	48	48.1	121	85	0.72	0.005	1.970 ± 0.038	0.18352 ± 0.00129	1.970 ± 0.038	0.18348 ± 0.00134	2646 ± 41	2685 ± 12	1.4
2	45	45.1	69	42	0.62	0.005	2.004 ± 0.041	0.18352 ± 0.00168	2.004 ± 0.041	0.18348 ± 0.00169	2609 ± 43	2685 ± 15	2.8
2	15	15.1	111	69	0.64	0.861	1.940 ± 0.036	0.19124 ± 0.00121	1.957 ± 0.037	0.18356 ± 0.00173	2661 ± 41	2685 ± 16	0.9
2	60	60.1	168	100	0.62	–0.056	1.970 ± 0.037	0.18323 ± 0.00107	1.969 ± 0.037	0.18373 ± 0.00111	2647 ± 41	2687 ± 10	1.5
2	34	34.1	163	89	0.56	–0.013	1.952 ± 0.036	0.18365 ± 0.00105	1.952 ± 0.036	0.18376 ± 0.00106	2667 ± 40	2687 ± 9	0.8
2	28	28.1	100	80	0.82	–0.035	2.020 ± 0.038	0.18349 ± 0.00134	2.019 ± 0.038	0.18381 ± 0.00134	2593 ± 41	2688 ± 12	3.5
2	39	39.1	129	31	0.25	–0.038	1.939 ± 0.037	0.18405 ± 0.00123	1.938 ± 0.037	0.18439 ± 0.00124	2682 ± 42	2693 ± 11	0.4
2	33	33.1	61	39	0.65	0.050	1.930 ± 0.039	0.18504 ± 0.00168	1.931 ± 0.039	0.18459 ± 0.00179	2690 ± 45	2695 ± 16	0.2
2	27	27.1	391	205	0.54	0.091	2.070 ± 0.036	0.18627 ± 0.00067	2.072 ± 0.036	0.18546 ± 0.00071	2539 ± 37	2702 ± 6	6.1
2	24	24.1	87	65	0.77	0.091	1.964 ± 0.040	0.18680 ± 0.00139	1.965 ± 0.040	0.18599 ± 0.00151	2652 ± 44	2707 ± 13	2.0
2	42	42.1	136	68	0.52	–0.017	1.916 ± 0.036	0.18590 ± 0.00119	1.916 ± 0.036	0.18605 ± 0.00119	2708 ± 41	2708 ± 11	0.0
2	3	3.1	51	17	0.35	0.781	1.875 ± 0.042	0.19303 ± 0.00198	1.890 ± 0.043	0.18606 ± 0.00292	2737 ± 51	2708 ± 26	–1.1
2	16	16.1	435	234	0.55	0.014	1.953 ± 0.036	0.18644 ± 0.00062	1.953 ± 0.036	0.18631 ± 0.00063	2665 ± 40	2710 ± 6	1.7
2	59	59.1	49	16	0.34	0.445	2.103 ± 0.047	0.19095 ± 0.00205	2.113 ± 0.048	0.18698 ± 0.00274	2498 ± 47	2716 ± 24	8.0
2	55	55.1	173	67	0.40	0.001	1.971 ± 0.036	0.18765 ± 0.00103	1.971 ± 0.036	0.18764 ± 0.00103	2646 ± 40	2722 ± 9	2.8
2	52	52.1	64	31	0.49	0.144	1.956 ± 0.042	0.18893 ± 0.00194	1.959 ± 0.042	0.18765 ± 0.00216	2658 ± 47	2722 ± 19	2.3
2	44	44.1	69	25	0.38	–0.006	1.804 ± 0.039	0.18773 ± 0.00160	1.803 ± 0.039	0.18779 ± 0.00160	2844 ± 49	2723 ± 14	–4.4
2	40	40.1	117	76	0.67	0.000	1.989 ± 0.038	0.18784 ± 0.00131	1.989 ± 0.038	0.18784 ± 0.00131	2626 ± 41	2723 ± 12	3.6
2	5	5.1	233	63	0.28	0.040	1.947 ± 0.035	0.18821 ± 0.00087	1.948 ± 0.035	0.18785 ± 0.00090	2671 ± 39	2723 ± 8	1.9
2	21	21.1	162	63	0.40	0.007	1.916 ± 0.035	0.18872 ± 0.00102	1.916 ± 0.035	0.18866 ± 0.00106	2707 ± 40	2730 ± 9	0.9

Table 1. (continued)**e**

Grp no.	*Spot no.*	*Grain .spot*	^{238}U *(ppm)*	^{232}Th *(ppm)*	$^{232}Th/^{238}U$	*f204 (%)*	$^{238}U/^{206}Pb$ ±1σ	$^{207}Pb/^{206}Pb$ ±1σ	$^{238}U/^{206}Pb^*$ ±1σ	$^{207}Pb^*/^{206}Pb^*$ ±1σ	$^{238}U/^{206}Pb^*$ *date (Ma)* ±1σ	$^{207}Pb^*/^{206}Pb^*$ *date (Ma)* ±1σ	*Disc (%)*
2	6	6.1	133	55	0.43	–0.002	1.947 ± 0.036	0.18909 ± 0.00116	1.947 ± 0.036	0.18911 ± 0.00116	2672 ± 41	2734 ± 10	2.3
2	22	22.1	120	83	0.71	0.005	1.921 ± 0.036	0.18949 ± 0.00118	1.921 ± 0.036	0.18944 ± 0.00119	2701 ± 41	2737 ± 10	1.3
2	29	29.1	99	41	0.43	0.037	1.908 ± 0.036	0.19223 ± 0.00131	1.909 ± 0.036	0.19190 ± 0.00135	2716 ± 42	2758 ± 12	1.5
2	41	41.1	207	55	0.28	0.003	1.875 ± 0.034	0.19444 ± 0.00100	1.875 ± 0.034	0.19441 ± 0.00101	2755 ± 41	2780 ± 9	0.9
2	58	58.1	547	299	0.56	0.067	2.032 ± 0.036	0.19570 ± 0.00139	2.034 ± 0.036	0.19510 ± 0.00140	2578 ± 37	2786 ± 12	7.5
2	10	10.1	797	48	0.06	0.038	2.003 ± 0.035	0.19828 ± 0.00050	2.004 ± 0.035	0.19794 ± 0.00052	2610 ± 37	2809 ± 4	7.1
2	37	37.1	16	7	0.48	–0.810	2.061 ± 0.068	0.19489 ± 0.00371	2.044 ± 0.068	0.20210 ± 0.00620	2567 ± 71	2843 ± 50	9.7
2	14	14.1	520	122	0.24	0.009	1.803 ± 0.032	0.21539 ± 0.00060	1.803 ± 0.032	0.21531 ± 0.00061	2844 ± 40	2946 ± 5	3.5
2	56	56.1	229	174	0.78	0.005	1.759 ± 0.032	0.21695 ± 0.00093	1.759 ± 0.032	0.21690 ± 0.00093	2902 ± 43	2958 ± 7	1.9
2	17	17.1	350	237	0.70	–0.013	1.738 ± 0.031	0.21719 ± 0.00071	1.737 ± 0.031	0.21730 ± 0.00071	2931 ± 42	2961 ± 5	1.0
2	8	8.1	352	151	0.44	0.017	1.534 ± 0.027	0.26663 ± 0.00081	1.534 ± 0.027	0.26649 ± 0.00082	3236 ± 45	3285 ± 5	1.5
2	4	4.1	721	305	0.44	0.040	1.608 ± 0.028	0.29841 ± 0.00243	1.609 ± 0.028	0.29807 ± 0.00243	3116 ± 43	3460 ± 13	9.9
3	61	61.1	1010	242	0.25	0.181	2.164 ± 0.038	0.16359 ± 0.00361	2.168 ± 0.038	0.16198 ± 0.00363	2445 ± 35	2476 ± 38	1.3
3	1	1.1	880	364	0.43	0.211	2.316 ± 0.040	0.16073 ± 0.00047	2.321 ± 0.040	0.15885 ± 0.00054	2310 ± 34	2443 ± 6	5.5
3	23	23.1	944	500	0.55	0.216	2.342 ± 0.040	0.15019 ± 0.00041	2.347 ± 0.040	0.14827 ± 0.00048	2288 ± 33	2326 ± 6	1.7
3	49	49.1	1333	627	0.49	0.190	2.690 ± 0.046	0.14006 ± 0.00042	2.695 ± 0.046	0.13838 ± 0.00050	2034 ± 30	2207 ± 6	7.8
3	53	53.1	2279	562	0.25	0.168	2.525 ± 0.043	0.13866 ± 0.00033	2.529 ± 0.043	0.13717 ± 0.00038	2148 ± 31	2192 ± 5	2.0
3	43	43.1	1995	652	0.34	0.264	2.801 ± 0.048	0.12853 ± 0.00084	2.808 ± 0.048	0.12621 ± 0.00091	1964 ± 29	2046 ± 13	4.0
3	47	47.1	2665	53	0.02	0.462	2.973 ± 0.051	0.12137 ± 0.00029	2.986 ± 0.051	0.11732 ± 0.00046	1862 ± 28	1916 ± 7	2.8
3	67	9.2	8416	1661	0.20	0.222	2.899 ± 0.049	0.11339 ± 0.00021	2.906 ± 0.049	0.11145 ± 0.00026	1907 ± 28	1823 ± 4	–4.6
3	9	9.1	7922	1503	0.20	0.223	3.027 ± 0.051	0.10873 ± 0.00047	3.033 ± 0.052	0.10679 ± 0.00050	1837 ± 27	1745 ± 9	–5.2
D	57	57.1	780	303	0.40	0.095	2.417 ± 0.042	0.16457 ± 0.00050	2.419 ± 0.042	0.16372 ± 0.00053	2230 ± 33	2494 ± 5	10.6
D	50	50.1	181	68	0.39	0.068	2.212 ± 0.041	0.18443 ± 0.00118	2.214 ± 0.041	0.18382 ± 0.00123	2403 ± 37	2688 ± 11	10.6
D	7	7.1	682	284	0.43	0.031	2.489 ± 0.043	0.16646 ± 0.00105	2.490 ± 0.043	0.16618 ± 0.00107	2176 ± 32	2520 ± 11	13.6

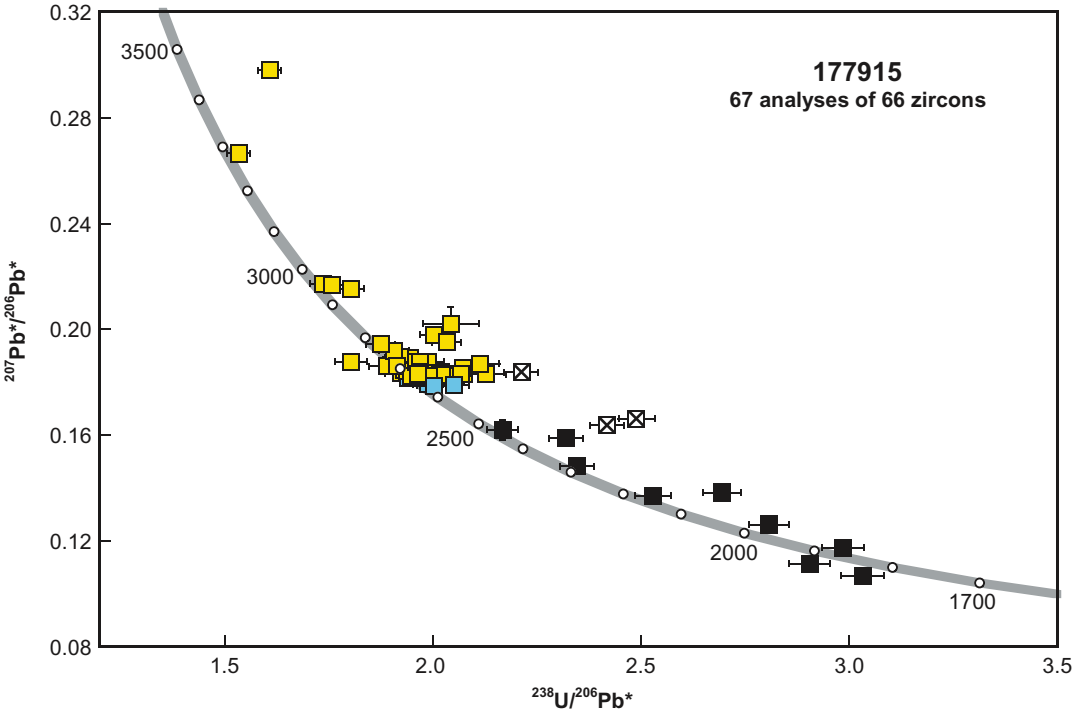

Figure 2. **U–Pb analytical data for sample 177915: metasandstone, Randalls Mine. Blue squares indicate Group 1 (youngest detrital zircons); yellow squares indicate Group 2 (older detrital zircons); black squares indicate Group 3 (radiogenic Pb loss); crossed white squares indicate ungrouped analyses (discordance >10%)**

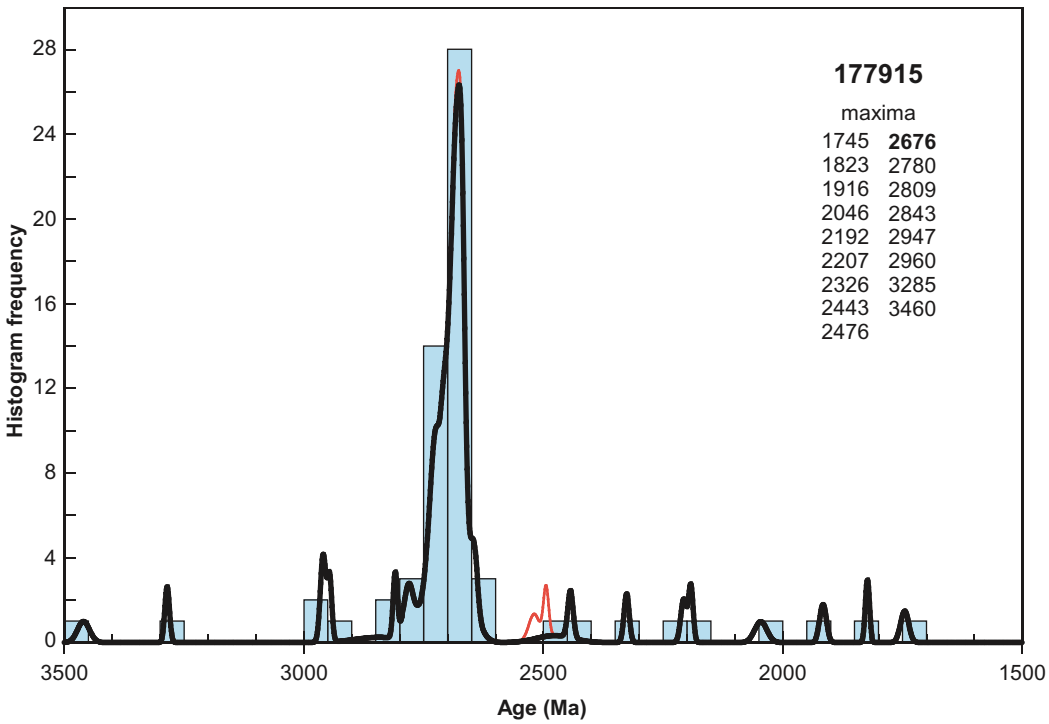

Figure 3. **Probability density plot and histogram for sample 177915: metasandstone, Randalls Mine. Heavy curve, maxima values, and frequency histogram (bin width 50 Ma) include only data with discordance <10% (64 analyses of 63 zircons). Lighter curve includes all data (67 analyses of 66 zircons)**